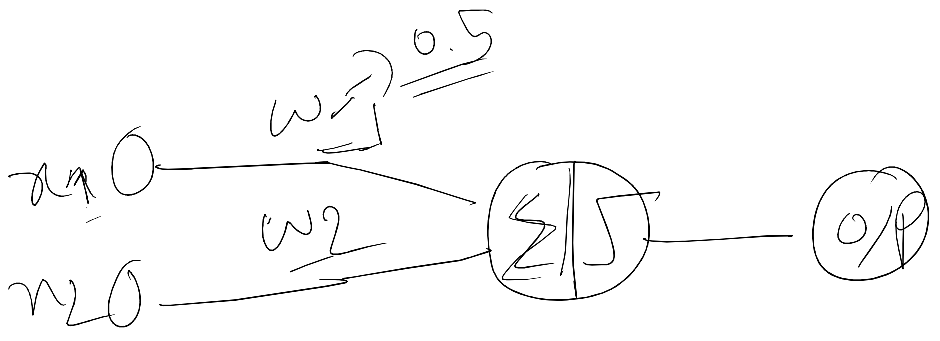
Weight Updation Formula

Monday, January 26, 2026 10:00 PM



weight updation formula:

$$w_{new} = w_{old} + \eta(\underbrace{y-\hat{y}}_{error}) \times x_i$$

η - Learning rate
0 - 1
y = Actual/target
$\hat{y}$ = prediction
x_i = input.

Gradient descent

special case,

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$$

Case:01 : y = 1 $\hat{y}$ = 0   error = 1

$$w_{new} = w_{old} + \eta(\underbrace{y-\hat{y}}_{error}) \times x_i$$

$$\underbrace{\qquad}_{\text{error}}$$

$$\Rightarrow \quad \underline{w}_{new} = w_{old} + \eta x_i$$

Case: 02: $y = 0 \quad \hat{y} = 1$, error $= -1$

$$w_{new} = w_{old} + \eta \underbrace{\overset{-1}{(y - \hat{y})}}_{\text{error}} \times \underline{x_i}$$

$$w_{new} = w_{old} - \eta \times x_i$$

Case: 03: $y = 0 \quad \hat{y} = 0$ | $y = 1$, $\hat{y} = 1$

error $= 0$

$$w_{new} = \underline{w_{old}} + \overbrace{\eta\underbrace{(y-\hat{y})}_{\text{error}}}^{0} \times \underline{x_i}$$

$$\boxed{w_{new} = w_{old}}$$